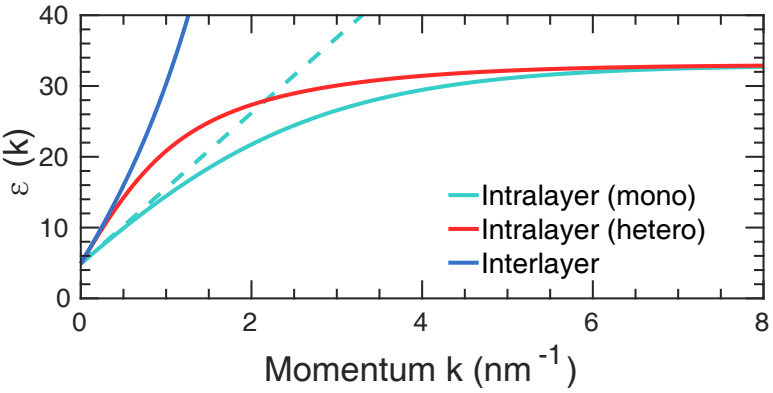
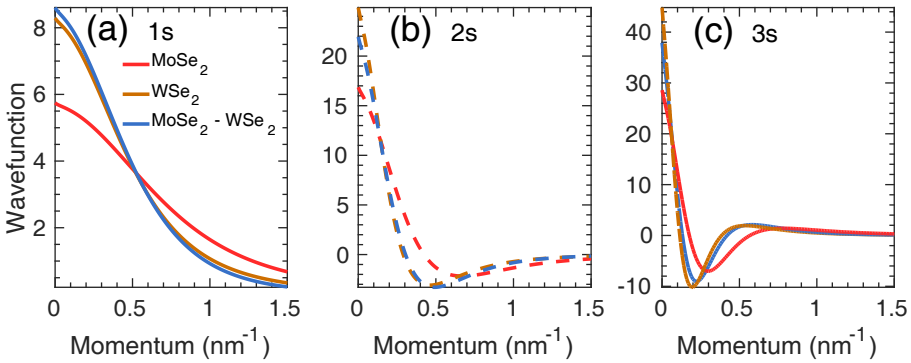
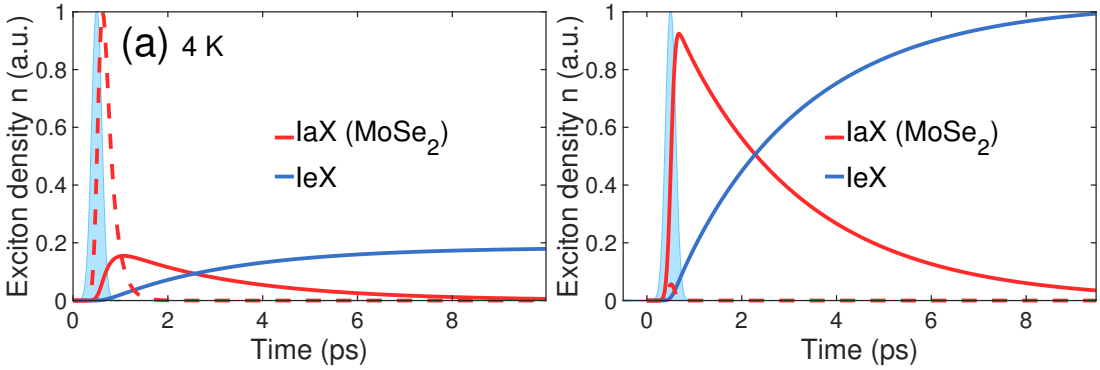


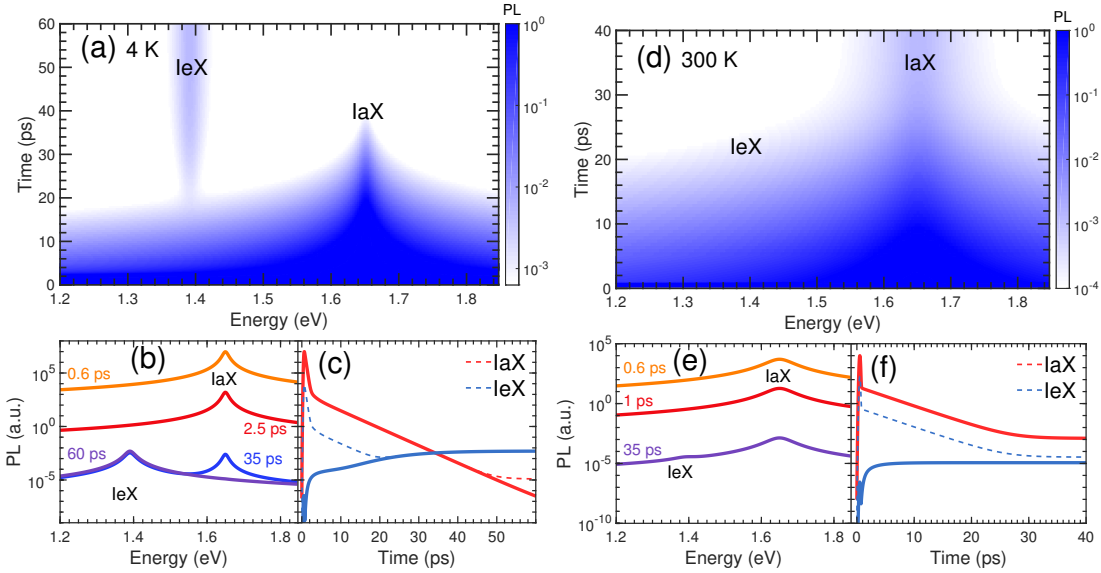
Supplementary Figure 1. VdW heterostructure dielectric functions. The dielectric function $\varepsilon(q)$ for the intralayer case in a monolayer, within a heterostructure and the interlayer case. The Keldysh limit describes the dielectric function after linearization and assuming that the dielectric constants are significantly larger than those of the surrounding media.



Supplementary Figure 2. Exciton wavefunctions. Wavefunctions of the lowest exciton states (1s (a), 2s (b), and 3s (c)) for the $MoSe_2$ and WSe_2 intralayer excitons and the corresponding $MoSe_2$-WSe_2 interlayer exciton, respectively.



Supplementary Figure 3. Momentum-integrated exciton dynamics. Evolution of the $MoSe_2$ intralayer (IaX) and interlayer (IeX) 1s exciton densities n after resonant optical excitation of the $MoSe_2$ layer at (a) 4 K and (b) 300 K (b). The optical pulse is denoted by the blue-shaded Gaussian. The dashed line represents coherent excitons in the $MoSe_2$ layer.



Supplementary Figure 4. **Time- and energy-resolved photoluminescence.** PL of the investigated heterostructure at 4 K (a-c) and 300 K (d-f) plotted (a, d) over time and energy (logarithmic) and showing snapshots taken at specific (b, e) times and (c, f) energies. The chosen energies correspond to the position of the interlayer (IeX) and $MoSe_2$ intralayer (IaX) exciton resonance, respectively. The dashed lines in (c, f) represent the signals stemming from the inter- or intralayer emission only, while the dashed lines also include the spectral overlap with the respective other peak.

Table of parameters

	$\hbar\omega^{\mathrm{LO}}$(meV)	$\hbar\omega^{\mathrm{TO}}$(meV)	$v^{\mathrm{LA}}\left(10^5\,\mathrm{cm\ s^{-1}}\right)$	$v^{\mathrm{TA}}\left(10^5\,\mathrm{cm\ s^{-1}}\right)$
$MoSe_2$	36.1	36.6	4.1	4.1
WSe_2	30.5	30.8	3.3	3.3
	$D_0^c\left(10^8\,\mathrm{eV\ cm^{-1}}\right)$	$D_0^v\left(10^8\,\mathrm{eV\ cm^{-1}}\right)$	D_1^c(eV)	D_1^v(eV)
$MoSe_2$	5.2	4.9	3.4	2.8
WSe_2	2.3	3.1	3.2	2.1
	d(nm)	ε	$m_{\mathrm{e}}(m_0)$	$m_{\mathrm{h}}(m_0)$
$MoSe_2$	0.645	16.5	0.50	0.60
WSe_2	0.645	15.1	0.29	0.36

Supplementary Table I. Table containing parameters used in the calculations. Their meaning can be found in the text.

SUPPLEMENTARY NOTE 1. EFFECTIVE 2D COULOMB POTENTIAL

To derive an effective 2D Coulomb potential we model the heterostructure as two aligned homogeneous slabs with the thicknesses $d_{1,2}$, infinite reach in the in-plane direction, and the dielectric constants $\varepsilon_{l_{1,2}}$. They are surrounded by dielectric media characterized by $\varepsilon_{1,2}$ and separated by a distance R, in which the medium has the permittivity ε_R. In the middle of layer 2 a point charge e_0 is placed. By performing the Fourier transforming in the in-plane direction, the Poisson equation can then be expressed as

$$\left(-k^2+\frac{\partial^2}{\partial z^2}\right)V(k,z)=-\frac{e_0^2}{\varepsilon_0\varepsilon_{l_2}}\delta(z). \tag{1}$$

Here, k is the in-plane momentum and $V(k,z)$ is the Coulomb potential. We search for a Coulomb potential that can be expressed as $V_k^{ll'}$, where the indices l,l' describe the layer of the two charges involved. By solving Eq. (1) within the two slabs and neglecting the in-layer z dependence, the solution gives us access to $V_k^{l_2l_1}$ and $V_k^{l_2l_2}$. By performing these steps we find that

$$V_k^{ll'}=\frac{e_0^2}{k\varepsilon_0\varepsilon^{ll'}(k)}, \tag{2}$$

where $\varepsilon^{ll'}(k)$ is the dielectric function of the system. For the interlayer case it has the form

$$\varepsilon^{l_2l_1}(k)=\cosh(kR)\frac{\cosh(kd_1)\cosh(kd_2)}{\cosh\left(k\frac{d_1}{2}\right)\cosh\left(k\frac{d_2}{2}\right)\left(1+\frac{\varepsilon_1}{\varepsilon_{l_1}}\tanh\left(k\frac{d_1}{2}\right)\right)\left(1+\frac{\varepsilon_2}{\varepsilon_{l_2}}\tanh\left(k\frac{d_2}{2}\right)\right)}f(k), \tag{3}$$

with

$$\begin{aligned}f(k)=&\left((\varepsilon_1+\varepsilon_2)+\left(\frac{\varepsilon_1\varepsilon_2}{\varepsilon_R}+\varepsilon_R\right)\tanh(kR)\right)\\&+\left(\left(\varepsilon_{l_1}+\frac{\varepsilon_1\varepsilon_2}{\varepsilon_{l_1}}\right)+\left(\frac{\varepsilon_{l_1}\varepsilon_2}{\varepsilon_R}+\frac{\varepsilon_1\varepsilon_R}{\varepsilon_{l_1}}\right)\tanh(kR)\right)\tanh(kd_1)\\&+\left(\left(\varepsilon_{l_2}+\frac{\varepsilon_1\varepsilon_2}{\varepsilon_{l_2}}\right)+\left(\frac{\varepsilon_{l_2}\varepsilon_1}{\varepsilon_R}+\frac{\varepsilon_2\varepsilon_R}{\varepsilon_{l_2}}\right)\tanh(kR)\right)\tanh(kd_2)\\&+\left(\left(\frac{\varepsilon_{l_2}}{\varepsilon_{l_1}}\varepsilon_1+\frac{\varepsilon_{l_1}}{\varepsilon_{l_2}}\varepsilon_2\right)+\left(\frac{\varepsilon_{l_1}\varepsilon_{l_2}}{\varepsilon_R}+\frac{\varepsilon_1\varepsilon_2}{\varepsilon_{l_1}\varepsilon_{l_2}}\varepsilon_R\right)\tanh(kR)\right)\tanh(kd_1)\tanh(kd_2).\end{aligned} \tag{4}$$

This expression diverges for $R\to\infty$ and the potential vanishes. The same result is obtained for $\varepsilon_R\to\infty$. In the limit $R\to 0$ it reduces to $(\varepsilon_1+\varepsilon_2)+k(\varepsilon_{l_1}d_1+\varepsilon_{l_2}d_2)$ after linearizing and assuming $\varepsilon_{1,2,R}\ll\varepsilon_{l_{1,2}}$. This is the Keldysh dielectric function for a single layer of the thickness d_1+d_2 and a dielectric constant $\frac{\varepsilon_{l_1}d_1+\varepsilon_{l_2}d_2}{d_1+d_2}$. Finally the expression also diverges for $k\to\infty$. To motivate this we consider the simpler case of two particles in a vacuum,

separated by $\sqrt{r^2+z^2}$, where $\mathbf{r}$ and z are in- and out-of-plane coordinates in an arbitrary coordinate system. After Fourier transforming $\mathbf{r}$ the corresponding Coulomb potential becomes $V(k,z) \propto \frac{1}{k}\int_0^\infty ds \frac{s}{\sqrt{s^2+k^2z^2}} J_0(s)$, where $J_0(s)$ is the zeroth order Bessel function of the first kind. This can be interpreted as the 2D Coulomb potential with an effective screening $1/\left(\int_0^\infty ds \frac{s}{\sqrt{s^2+k^2z^2}} J_0(s)\right)$, a screening that diverges for $k \to \infty$.

For the intralayer case the result reads

$$\begin{aligned}\varepsilon^{l_2 l_2}(k) = &\frac{\cosh(kd_2)}{\cosh^2\left(k\frac{d_2}{2}\right)\left(1+\tanh\left(k\frac{d_2}{2}\right)\frac{\varepsilon_2}{\varepsilon_{l_2}}\right)} f(k)/ \\ &/\left[\left(1+\frac{\varepsilon_1}{\varepsilon_R}\tanh(kR)\right)+\left(\frac{\varepsilon_1}{\varepsilon_{l_1}}+\frac{\varepsilon_{l_1}}{\varepsilon_R}\tanh(kR)\right)\tanh(kd_1)\right. \\ &\left.+\left(\frac{\varepsilon_1}{\varepsilon_{l_2}}+\frac{\varepsilon_R}{\varepsilon_{l_2}}\tanh(kR)\right)\tanh\left(k\frac{d_2}{2}\right)+\left(\frac{\varepsilon_{l_1}}{\varepsilon_{l_2}}+\frac{\varepsilon_1\varepsilon_R}{\varepsilon_{l_1}\varepsilon_{l_2}}\tanh(kR)\right)\tanh(kd_1)\tanh\left(k\frac{d_2}{2}\right)\right].\end{aligned}$$

For $k \to 0$, we obtain $\varepsilon_1+\varepsilon_2$, which is consistent with the Keldysh case, while $k \to \infty$ results in ε_{l_2}. When $\varepsilon_1 = \varepsilon_{l_1} = \varepsilon_R$ the R dependence completely vanishes, and for $\varepsilon_{1,2} \ll \varepsilon_{l_2}$ results in the ordinary Keldysh dielectric function after linearization. The limit $R \to \infty$ results in the same, but with ε_R taking the role of ε_1.

Supplementary Figure 1 shows a comparison of the dielectric function $\varepsilon(q)$ for the intralayer case in a monolayer, within a heterostructure and the interlayer. As parameters we have used $d_1 = d_2 = 0.645\,\text{nm}$ and $R = 0\,\text{nm}$. For the dielectric constants, we have assumed $\varepsilon_{1,R} = 1$ and $\varepsilon_2 = 3.9$ representing air and SiO_2, as well as $\varepsilon_{l_1} = 15.1$ and $\varepsilon_{l_2} = 16.5$ for WSe_2 and $MoSe_2$.

SUPPLEMENTARY NOTE 2. EXCITONIC WAVEFUNCTIONS

The excitonic wavefunctions are obtained through solving the Wannier equations, cf. Supplementary Figure 2. A general trend that can be observed is that a lower binding energy leads to sharper excitonic wavefunctions in the momentum space corresponding to more delocalized excitonic states in the real space. A consequence of this is that interactions involving interlayer excitons will favor scattering events with lower momentum transfers.

SUPPLEMENTARY NOTE 3. TEMPERATURE DEPENDENCE OF EXCITON DYNAMICS

While in the main text, intra- and interlayer exciton dynamics and photoluminescence has been shown at 77 K, here we present the momentum integrated dynamics calculated at 4 K and 300 K. Supplementary Figure 3 shows how the coherence lifetime drastically increases at very low temperatures, which can be seen by comparing the red dashed lines in Supplementary Figure 3(a) and Supplementary Figure 3(b). This behavior can be traced back to the reduced efficiency of exciton-phonon scattering, which also explains the less pronounced polarization-to-population transfer, i.e. the formation of incoherent excitons (red solid lines).

This has a direct impact on the photoluminescence, which is presented in Supplementary Figure 4. At 4 K (Supplementary Figure 4(a - c)) exciton-phonon scattering becomes significantly less efficient relative to the interlayer tunneling. As a result, there is a non-negligible delay until interlayer excitons scatter into the light cone and become visible in PL. As a result, the IeX emission does not saturate until it surpasses the IaX resonance at 35 ps (blue line in Supplementary Figure 4(b)). Instead it slowly increases until it reaches equilibrium at approximately 60 ps (purple line in Supplementary Figure 4(b)). This increase can be seen in more detail in Supplementary Figures 4(a, c). At 300 K (Supplementary Figures 4(d - f)) the IeX resonance can barely be seen due to the broadening of the intralayer exciton transition. This can be well seen in Supplementary Figure 4(f). In contrast to 4 K and 77 K, the total emission at the IeX energy including off-resonant IaX emission (dashed blue line) exceeds the emission from IeX alone (solid blue line).

SUPPLEMENTARY NOTE 4. DFT CALCULATIONS

The structure consisting of a $MoSe_2$ and a WSe_2 layer in a 2H configuration was relaxed in the software VASP [1] with both PBE and vdW-DF-cx until the maximal force was 5 meV/Å. The plane wave cutoff energy was 300 eV and the Brillouin zone was sampled with a $15 \times 15 \times 5$ **k**-point mesh. The PBE bilayer (with in-plane lattice constant 3.32 Å) was then isolated and a vacuum of 10 Å was applied to the bilayer. One monolayer was then shifted such

that the Mo-W distance matched the value obtained from the relaxation with the vdW-DF-cx functional (6.45 Å). The bilayer system was then split up into two configurations consisting of a single layer. The wavefunctions of the separate layers were then computed on a real space grid with GPAW [2] using norm conserving potentials and a grid spacing of 0.2 Å^{-1}. The **k**-point mesh for the self-consistent calculation of the electron density was $33 \times 33 \times 1$. The wavefunctions were then computed on a radial grid centered at K that extended 0.4 Å^{-1}. The radial grid consisted of 25 radial points (including K) and 40 angular points. The value of the overlap integral (for a smaller k-point set) was checked with a reduced grid spacing of 0.1 Å^{-1}.

SUPPLEMENTARY NOTE 5. LUMINESCENCE BLOCH EQUATIONS

The luminescence Bloch equations shown in the main text can be more explicitly expressed as

$$\dot{P}^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}} = \frac{1}{i\hbar} E^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}} P^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}} + i\Omega^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}} - \left(\gamma^{l_\mathrm{h}l_\mathrm{e}}_\mathrm{r} + \frac{1}{2}\sum_{\mathbf{Q}'} \Gamma^{l_\mathrm{h}l_\mathrm{e}}_{\mathrm{P},\mathbf{Q}\mathbf{Q}'}\right) P^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}} \tag{5}$$

$$\begin{aligned}\dot{N}^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}} = &\sum_{\mathbf{Q}'} \Gamma^{l_\mathrm{h}l_\mathrm{e}}_{\mathrm{P},\mathbf{Q}'\mathbf{Q}} |P^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}'}|^2 - 2\gamma^{l_\mathrm{h}l_\mathrm{e}}_\mathrm{r}\delta_{\mathbf{Q},0} N^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}} + \sum_{\mathbf{Q}'} (\Gamma^{l_\mathrm{h}l_\mathrm{e}}_{\mathrm{P},\mathbf{Q}'\mathbf{Q}} N^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}'} - \Gamma^{l_\mathrm{h}l_\mathrm{e}}_{\mathrm{P},\mathbf{Q}\mathbf{Q}'} N^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}}) \\ &+ \sum_{\mathbf{Q}'l'_\mathrm{h}l'_\mathrm{e}} \Gamma^{l_\mathrm{h}l_\mathrm{e}l'_\mathrm{h}l'_\mathrm{e}}_{\mathrm{T},\mathbf{Q}\mathbf{Q}'} (N^{l'_\mathrm{h}l'_\mathrm{e}}_{\mathbf{Q}'} - N^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}}).\end{aligned} \tag{6}$$

The first two terms describing the dynamics of the exciton polarization $P^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}}(t)$ with hole (electron) in the layer l_h (l_e) are determined by the excitonic dispersion $E^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}}$ and the Rabi frequency $\Omega^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}}(t) = i\frac{e_0}{m_0}\sum_{\mathbf{q}} \varphi^{*l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}} \mathbf{M}^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}} \cdot \mathbf{A}(t)$. Here, e_0 and m_0 are the elementary charge and electronic rest mass, $\varphi^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}}$ is the excitonic wavefunction, $\mathbf{M}^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}}$ the optical matrix element describing optical interband transitions [3, 4] and $\mathbf{A}(t)$ is the driving optical pump pulse. The decay processes stem from the radiative damping is described by the rate $\gamma^{l_\mathrm{h}l_\mathrm{e}}_\mathrm{r} = \frac{\hbar^2 c_0 \mu_0}{\omega^{l_\mathrm{h}l_\mathrm{e}} n}\left|\sum_{\mathbf{q}} \varphi^{*l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}} M^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}}\right|^2$, where $\frac{c_0}{n}$ is the speed of light in the substrate material, μ_0 the vacuum permeability and $\omega^{l_\mathrm{h}l_\mathrm{e}}$ the resonant photon frequency [4]. The exciton-phonon interaction is determined by the rate $\Gamma^{l_\mathrm{h}l_\mathrm{e}}_{\mathrm{P},\mathbf{Q},\mathbf{Q}'} = \frac{2\pi}{\hbar}\sum_{l\alpha\pm} |G^{l_\mathrm{h}l_\mathrm{e}l\alpha}_{\mathbf{Q}-\mathbf{Q}'}|^2 \hat{n}^{l\alpha\pm}_{\mathbf{Q}-\mathbf{Q}'} \delta(\Delta E^{l_\mathrm{h}l_\mathrm{e}l\alpha\pm}_{\mathbf{Q},\mathbf{Q}'})$, where the scattering cross section is given by $G^{l_\mathrm{h}l_\mathrm{e}l\alpha}_{\mathbf{K}} = \sum_{\mathbf{q}} \varphi^{*l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}} \left(\varphi^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}+\beta^{l_\mathrm{h}l_\mathrm{e}}\mathbf{K}} g^{c,l_\mathrm{e}\alpha}_{\mathbf{K}} \delta^{l_\mathrm{e}l} - \varphi^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}-\alpha^{l_\mathrm{h}l_\mathrm{e}}\mathbf{K}} g^{v,l_\mathrm{h}\alpha}_{\mathbf{K}} \delta^{l_\mathrm{h}l}\right)$. Here, $\alpha^{l_\mathrm{h}l_\mathrm{e}} = \frac{m^{l_\mathrm{e}}_\mathrm{e}}{m^{l_\mathrm{h}}_\mathrm{h}+m^{l_\mathrm{e}}_\mathrm{e}}$ and $\beta^{l_\mathrm{h}l_\mathrm{e}} = \frac{m^{l_\mathrm{h}}_\mathrm{h}}{m^{l_\mathrm{h}}_\mathrm{h}+m^{l_\mathrm{e}}_\mathrm{e}}$ and $g^{\lambda,l\alpha}_{\mathbf{K}}$ is the electron-phonon matrix element for band λ (with c and v denoting the conduction and valence bands), layer l and phonon mode and momentum α and $\mathbf{K}$. Furthermore, $\Delta E^{l_\mathrm{h}l_\mathrm{e}l\alpha\pm}_{\mathbf{Q},\mathbf{Q}'} = E^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}'} - E^{l_\mathrm{h}l_\mathrm{h}}_{\mathbf{Q}} \pm \hbar\omega^{l\alpha}_{\mathbf{Q}-\mathbf{Q}'}$ describes the energy difference when absorbing ($-$) or emitting ($+$) a phonon with the energy $\hbar\omega^{l\alpha}_{\mathbf{Q}-\mathbf{Q}'}$. Finally, $\hat{n}^{l\alpha\pm}_{\mathbf{Q}-\mathbf{Q}'} = \frac{1}{2} \pm \frac{1}{2} + n^{l\alpha}_{\mathbf{Q}-\mathbf{Q}'}$, where the phonon occupation $n^{l\alpha}_K = \frac{1}{\exp\left(\frac{\hbar\omega^{l\alpha}_{\mathbf{K}}}{k_\mathrm{B}T}\right)-1}$ with Boltzmann constant k_B and temperature T is treated as a Bose-Einstein distribution [4, 5]. The electron-phonon matrix element can be expressed as $g^{\lambda,l\alpha}_{\mathbf{K}} = \sqrt{\frac{\hbar}{2M^l\omega^{l\alpha}_{\mathbf{K}}}} V^{\lambda,l\alpha}_{\mathbf{K}}$, where M^l is the unit cell mass and $\omega^{l\alpha}_{\mathbf{K}}$ the phonon frequency. The deformation potential is taken as constant for optical phonons ($V^{\lambda,l\alpha}_{\mathbf{K}} = D^{\lambda,l\alpha}_0$) and linear for acoustic ($V^{\lambda,l\alpha}_{\mathbf{K}} = D^{\lambda,l\alpha}_1 K$). In the same way the phonon energy is constant for optical phonons and linear for acoustic ($\hbar\omega^{l\alpha}_{\mathbf{K}} = \hbar v^{l\alpha} K$ where $v^{l\alpha}$ is the sound velocity in layer l).

The dynamics of the incoherent exciton occupation $N^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}}(t)$ is determined by formation processes driven by phonon-assisted decay of the excitonic polarization, the radiative decay, and exciton-phonon scattering driving the excited system towards an equilibrium Boltzmann-like distribution [5, 6]. The final term in Eq. (6) describes resonant tunneling between different layers and causes the formation of interlayer excitons. It has an effective scattering rate $\Gamma^{l_\mathrm{h}l_\mathrm{e}l'_\mathrm{h}l'_\mathrm{e}}_{\mathrm{T},\mathbf{Q}\mathbf{Q}'} = \frac{2\pi}{\hbar}|\tilde{T}^{l_\mathrm{h}l_\mathrm{e}l'_\mathrm{h}l'_\mathrm{e}}_{\mathbf{Q}\mathbf{Q}'}|^2 \delta(E^{l'_\mathrm{h}l'_\mathrm{e}}_{\mathbf{Q}'} - E^{l_\mathrm{h}l_\mathrm{e}}_{\mathbf{Q}})$, where the exciton tunneling cross section is expressed as $\tilde{T}^{l_\mathrm{h}l_\mathrm{e}l'_\mathrm{h}l'_\mathrm{e}}_{\mathbf{Q}\mathbf{Q}'} = \sum_{\mathbf{q}} \left(\varphi^{*l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}+\beta^{l_\mathrm{h}l_\mathrm{e}}\mathbf{Q}} \varphi^{l'_\mathrm{h}l'_\mathrm{e}}_{\mathbf{q}+\beta^{l'_\mathrm{h}l'_\mathrm{e}}\mathbf{Q}'} T^{c,l_\mathrm{e}l'_\mathrm{e}}_{\mathbf{q}+\mathbf{Q},\mathbf{q}+\mathbf{Q}'} (1-\delta^{l_\mathrm{e}l'_\mathrm{e}})\delta^{l_\mathrm{h}l'_\mathrm{h}} - \varphi^{*l_\mathrm{h}l_\mathrm{e}}_{\mathbf{q}-\alpha^{l_\mathrm{h}l_\mathrm{e}}\mathbf{Q}} \varphi^{l'_\mathrm{h}l'_\mathrm{e}}_{\mathbf{q}-\alpha^{l'_\mathrm{h}l'_\mathrm{e}}\mathbf{Q}'} T^{*v,l_\mathrm{h}l'_\mathrm{h}}_{\mathbf{q}-\mathbf{Q},\mathbf{q}-\mathbf{Q}'} \delta^{l_\mathrm{e}l'_\mathrm{e}} (1-\delta^{l_\mathrm{h}l'_\mathrm{h}})\right)$. Here $T^{\lambda,ll'}_{\mathbf{K},\mathbf{K}'} = \frac{\sqrt{\pi} L_\mathrm{C}}{\left(1+\frac{|\mathbf{K}'-\mathbf{K}|^2 L_\mathrm{C}^2}{2}\right)^{\frac{3}{4}}} V_z \left\langle u^{\lambda l}_{\mathbf{K}} | u^{\lambda l'}_{\mathbf{K}} \right\rangle_\mathrm{uc}$ is the electron tunneling matrix element that has been discussed in the main text. The numerical values of the parameters used are given in Supplementary Table 1.

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